

Lakhdar Berache

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PROFILE

Software and AI engineer specialising in multi-agent LLM systems. One year in automotive AI, with Renault Korea's R&D teams and as a consultant in France, taking features from prototype to production under embedded and safety constraints. Looking to contribute to work that advances the technology and proves useful to the people who rely on it. Twice awarded in national and international innovation competitions, graduating in July 2027.

PROFESSIONAL EXPERIENCE

ArcherMind Technology (Nanjing) Co., Ltd.

Paris, France (remote)

Software & AI Engineer, contract

Mar 2026 – Present

- Co-developed Renault Korea's AI Orchestrator, the in-vehicle AI system presented at NextRise 2026 alongside Kakao Mobility, TMAP Mobility and Valeo.
- Delivered a client project end to end: technical design, stack selection, implementation, integration and production rollout.
- Build and evaluate multi-agent workflows (LangGraph, LangChain) for intent understanding, context handling and vehicle-function orchestration through tool-calling agents.
- Work under embedded constraints: limited compute, offline tolerance, automotive HMI and functional-safety requirements.

Renault Group – Renault Korea R&D

Yongin, South Korea

Software & AI Engineer Intern

Sep 2025 – Feb 2026

- Initiated the AI Orchestrator that Renault later showcased at NextRise 2026.
- Prototyped and integrated AI features for infotainment and connected services: personalisation, in-car assistant, generated in-car content.
- Built a multi-agent conversational system with LangGraph, integrating MCP servers, a RAG pipeline over vehicle-manual knowledge, real-time TTS streaming and cloud messaging to mobile.
- Took features from prototype to tested components across connected services and embedded targets; reported monthly in English to director- and executive-level stakeholders.

D-Valentin

Paris, France (remote)

Co-Founder & AI Lead

Jul 2026 – Present

- Lead the software and AI architecture of a multi-agent analysis platform whose every output carries its source, date and confidence: orchestration framework, model selection and evaluation methodology. Venture born at Google's Build with Gemini XPRIZE.

EDUCATION

University of Technology of Troyes (UTT)

Troyes, France

Master of Engineering (Diplôme d'Ingénieur), Computer Science & Information Systems

2022 – 2027

- Specialisation in Informatics and Systems (ISI); integrated preparatory program 2022–2024.

Korea Advanced Institute of Science and Technology (KAIST)

Daejeon, South Korea

Exchange semester, Informatics and Artificial Intelligence, taught in English

Feb – Jun 2025

SELECTED PROJECTS & ACHIEVEMENTS

- Astyr** (*astyr.app*) – AI process optimisation for industrial operations, in production across automotive, marine and infrastructure assets. Started as ARC: **2nd place, Vultr track, RAISE Summit Hackathon Paris** (Jul 2026), among 6,000+ builders from 60 countries.
- Wallside** – retrieval-augmented assistant over a corpus of 165 vehicle manuals, 62,700 pages and 138,000 passages. Built the evaluation harness that measures retrieval accuracy with confidence intervals, and used it to raise document identification from 26% to **98%** at full corpus size through metadata routing over a hybrid retriever.
- SimuPont** – portable training simulator for the nuclear industry. **Public's Favourite Award and finalist**, Megawatt'UTT Challenge 26 (EDF / Fondation UTT); covered in *Le Parisien*.

SKILLS, CERTIFICATIONS & LANGUAGES

Programming Python, TypeScript / JavaScript, Java, SQL.

AI & ML LLM agents and multi-agent orchestration (LangGraph, LangChain, MCP); retrieval-augmented generation; NLP; computer vision; Hugging Face Transformers; scikit-learn.

Tools Docker, FastAPI, Flask, React, PostgreSQL, MySQL, MongoDB, AWS (Bedrock), Git, Agile, UML.

Certifications Getting Started with AWS Generative AI for Developers (AWS, 2025); Build RAG Applications (IBM, 2025).

Languages French (native), Arabic (bilingual), English (fluent), Korean (beginner).